

A Delphi-Based Framework for systems architecting of in-orbit exploration infrastructure for human exploration beyond Low Earth Orbit[☆]



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ABSTRACT

The current debate in the U.S. Human Spaceflight Program focuses on the development of the next generation of man-rated heavy lift launch vehicles. While launch vehicle systems are of critical importance for future exploration, a comprehensive analysis of the entire exploration infrastructure is required to avoid costly pitfalls at early stages of the design process. This paper addresses this need by presenting a Delphi-Based Systems Architecting Framework for integrated architectural analysis of future in-orbit infrastructure for human space exploration beyond Low Earth Orbit. The paper is structured in two parts.

The first part consists of an expert elicitation study to identify objectives for the in-space transportation infrastructure. The study was conducted between November 2011 and January 2012 with 15 senior experts involved in human spaceflight in the United States and Europe. The elicitation study included the formation of three expert panels representing exploration, science, and policy stakeholders engaged in a 3-round Delphi study. The rationale behind the Delphi approach, as imported from social science research, is discussed. Finally, a novel version of the Delphi method is presented and applied to technical decision-making and systems architecting in the context of human space exploration.

The second part of the paper describes a tradespace exploration study of in-orbit infrastructure coupled with a requirements definition exercise informed by expert elicitation. The uncertainties associated with technical requirements and stakeholder goals are explicitly considered in the analysis. The outcome of the expert elicitation process portrays an integrated view of perceived stakeholder needs within the human spaceflight community. Needs are subsequently converted into requirements and coupled to the system architectures of interest to analyze the correlation between exploration, science, and policy goals. Pareto analysis is used to identify architectures of interest for further consideration by decision-makers.

The paper closes with a summary of insights and develops a strategy for evolutionary development of the exploration infrastructure of the incoming decades. The most important result produced by this analysis is the identification of a critical irreducible ambiguity undermining value delivery for the in-space transportation infrastructure of the next three decades: destination choice. Consensus on destination is far from being reached by the community at large, with particular reference to exploration and policy stakeholders. The realization of this ambiguity is a call for NASA to promote an open forum on this topic, and to develop a strong case for policy makers to incentivize investments in the human spaceflight industry in the next decades.

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1. Introduction

While NASA has already selected the Space Launch System as the new man-rated launch vehicle [1,2], one important question that is still open is the definition of the remaining architectural elements of the transportation infrastructure.

This paper supports the architecting process of the future US human spaceflight program by implementing a Delphi-based Systems Architecting Framework (DB-SAF) [3] for comprehensive architecting of the in-orbit transportation infrastructure. DB-SAF supports the objective definition of design goals by identifying and characterizing ambiguities in value objectives through expert elicitation, allowing experts to negotiate with each other through an anonymous and interactive process, and view the impact of their preferences in the architectural tradespace. The analysis is structured in two parts: first, possible objectives for the infrastructure are explored through an expert elicitation approach, with the goal of identifying objectives and characterizing associated technical, scientific, and political uncertainties. The study involved experts from NASA, ESA, academia and industry. The second part of the paper performs a tradespace exploration based on value metrics developed through expert elicitation, developing recommendations on the architecting process. The analysis, therefore, investigates what efficient development strategies could be pursued for future in-space infrastructure, while does not investigate the fundamental rationale of human spaceflight, which has been investigated extensively in recent years by scholars in space policy [4–6].

The paper shows how DB-SAF can be used to characterize and mitigate ambiguity in system-level objectives, therefore being an effective decision-making tool supporting system architects in the design of large engineering infrastructures.

The remainder of the paper is structured as follows: Section 2 provides the motivations and historical context that categorize this study as relevant to the human spaceflight program; Section 3 describes formally the objectives of the study; Section 4 describes the approach used in this paper, namely the Delphi-Based Systems Architecting Framework (DB-SAF), and describes the results that have been obtained following its application to the in-space infrastructure architecting problem; Section 5 summarizes the findings of this paper for consideration by decision-makers involved in the development of the future in-space infrastructure for human exploration.

2. Motivations and context

This section provides the framing context that motivated this research, including relevant historic background on the US human spaceflight program. NASA is presently called to charter a path for its future plans for human exploration. In 2004, President Bush started the Constellation Program with the ambitious goal of returning astronauts to the surface of the Moon [7]. Following 6 years of development and a cumulative investment of \$9B billion USD [8], a Presidential panel was chartered with the purpose of assessing the status of the US Human

Spaceflight Program and providing recommendations to the White House for future development of the American manned spaceflight program (the “Augustine Committee”) [9]. The Committee found the US human spaceflight program to be “*on an unsustainable trajectory <...> perpetuating the perilous practice of pursuing goals that do not match allocated resources*” [9]. Following this review, the Obama Administration cancelled Constellation and the programs therein, such as the Ares I and Ares V launch vehicles [10]. Eventually, in July 2011 the Space Shuttle completed its last flight and transitioned to its decommissioning phase, marking the end of an era in human spaceflight [11].

In September 2011, NASA announced the development of the Space Launch System (SLS) as the next generation launch vehicle for future human exploration [12]. While most efforts have been focused on the development of the launch system, less effort seems to have been spent upfront on the remaining elements of the exploration infrastructure. This is partly due to political uncertainty on overall program objectives, which makes it difficult to commit to any hardware development that can constrain destination selection and other program objectives in successive development phases. Stakeholders do not share a unified vision on what constitutes value for the exploration enterprise. They have variegated perspectives on the intrinsic value of space exploration. Debates generate ambiguities that must be identified, characterized, and successively mitigated through careful strategy to define a robust transportation architecture.

3. Objectives

This paper characterizes and identifies ambiguities in system objectives for the in-orbit transportation infrastructure for human space exploration. It provides a characterization of the debate on “what is value” in future human spaceflight programs as perceived by a panel of experts. Based on this analysis, the paper enumerates feasible architectures for the in-space infrastructure and identifies architectures of interest while developing recommendations for further consideration by decision-makers. Architectures are filtered through proxy metrics for cost, schedule, and risk.

4. Approach and results

This section presents the approach that has been used for the analysis of in-orbit infrastructure presented in this paper, and the results that have been obtained by its implementation. The approach is called Delphi-Based Systems Architecting Framework (DB-SAF) [3], and its overview is shown in Fig. 1. DB-SAF is an iterative approach integrating expert elicitation and computational systems architecting. DB-SAF supports the definition of system architectures under ambiguous stakeholder objectives; that is when stakeholders have a plurality of contrasting views on what capabilities should be implemented in the system and what constitutes value for the architecture. DB-SAF is inspired by the Delphi method from social science research [13,14], and integrates an engineering Delphi approach with computational systems architecting techniques developed at MIT in recent years [15,16]. DB-SAF allows experts to be included in the

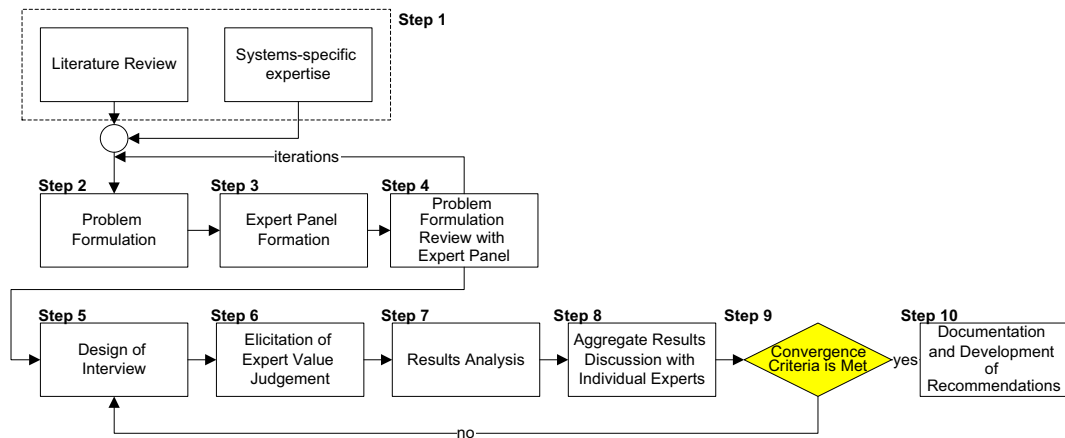


Fig. 1. Delphi-Based Systems Architecting Framework (DB-SAF).

definition of a systems architecting problem and to interact between each other in anonymous form in negotiating judgments on questions of interest – for instance, on what constitutes value for an exploration infrastructure. This approach is iterative, as experts go through several interview rounds until convergence criteria are met. An in-depth explanation of the DB-SAF methodology can be found in [3]. This section of the paper shows how DB-SAF has been implemented for the in-space exploration infrastructure systems architecting problem. In the sub-sections that follow, the step numbering shown in Fig. 1 is referenced.

4.1. Step 1 – literature review

The first step in DB-SAF is a literature search to inform the architecting study. In particular, the goal is to find evidence on how previous system architectures were evaluated in terms of value judgments: figures of merit, evaluation metrics, and so forth. The literature in human spaceflight mission architectures is very rich, as many proposals and mission architectures were studied in the last 50 years. The main references considered in the study presented in this paper are the major design efforts in this direction starting from 2000 with particular emphasis on NASA and MIT human spaceflight studies. Among others, this study considered findings from MIT-Draper's Concept Exploration and Refinement Exploration Systems Architecture Study [17], NASA's Exploration Systems Architecture Study (ESAS) [7,18], the Mars Design Reference Architecture 5.0 [19], and the NASA architectures for human missions on Near Earth Asteroids [20,21].

4.2. Step 2 – problem formulation

4.2.1. Identification of questions of interest

This paper aims to address the following questions of relevance to the systems architecting problem of in-space transportation infrastructure for human exploration beyond Low Earth Orbit:

- What destinations should we consider pursuing for the US human exploration program of the next 30 years?

- What types of mission (orbit/surface/short stay/long stay) should we consider as effective compromise between exploration value, science, and policy return?
- What are the main tradeoffs in exploration value, science, and policy return in architecting the in-space transportation infrastructure for human exploration? How do these metrics affect the architecting process as evaluated by performance, cost and risk metrics, and what is the impact of ambiguity in this context?

4.2.2. Goals identification

As the rationale for human space exploration is fairly diverse within stakeholders, there is no crisply defined goal on which stakeholders agree for required in-space infrastructure for manned spaceflight. For the purposes of this paper, we will consider a subset of the main Key Supporting Objectives (KSO) for human space exploration defined by the Global Exploration Roadmap (GER) of the International Space Exploration Coordination Group [22] relevant to the in-space transportation infrastructure architecture:

- | | |
|---|---------|
| - Extend Human Presence | [KSO 1] |
| - Develop Exploration Technologies and Capabilities | [KSO 2] |
| - Perform Science to Support Human Exploration | [KSO 3] |
| - Stimulate Economic Expansion | [KSO 4] |
| - Perform Space, Earth, and Applied Science | [KSO 5] |
| - Engage Public in Exploration | [KSO 6] |

The Global Exploration Roadmap derives objectives from these goals; however, it does not provide rationale on how these objectives are linked to each other and how should they be prioritized in the program. For instance, boil-off control and in-situ resource utilization are both objectives mapped to KSO 2 in GER. This paper addresses these questions concurrently and identifies interactions between technology investments, providing insight for future developments.

4.2.3. Functional decomposition

Functions are form-neutral statements derived from the goals as expressed by stakeholders, also referred to

as functional requirements. The in-space transportation infrastructure has been modeled with the following functional decomposition:

- Providing Habitation to astronauts at all phases during the length of the mission
- Transporting People and Cargo from Low Earth Orbit to the Destination

Two main sources of ambiguity that have been underlined: the length of the mission, which is determined by the chosen mission mode, the time spent at the destination and the choice of the destination itself; and destinations, which choice drives value delivery to exploration, science and policy stakeholders.

4.2.4. Requirements enumeration

Areas of potential ambiguity impacting definition of requirements include the following requirement variables:

- The choice of a **destination** (affecting value delivery to stakeholders in goals KSO 1, KSO 2, KSO 3, KSO 4, KSO 5, KSO 6)
- Characteristics associated with NEA destinations, such as **characteristic size** and **internal composition** (affecting value delivery to stakeholders in goals KSO 2, KSO 3, KSO 4, KSO 6)
- The **number of crew** in the mission (affecting value delivery to stakeholders in goals KSO 1, KSO 2, KSO 3, KSO 6)
- **Exploration time available at the destination** for exploration and science activities (affecting value delivery to stakeholders in goals KSO 1, KSO 2, KSO 3, KSO 4, KSO 5, KSO 6)
- The **time of flight of the mission, as a proxy measure of health risk** to astronauts due to exposure to the harsh radiation environment beyond Low-Earth Orbit (affecting value delivery to stakeholders in goals KSO 1 and KSO 2)

Table 1 shows the requirements options that have been formulated to assess ambiguities in more detail, specifying alternative options to be considered for each requirement variable. By considering all possible combinations of options from the table, the size of the unconstrained

requirements tradespace in this formulation is 21,600 possible requirement sets (unconstrained estimate). To allow comparison with existing architectural baselines, the analysis is focused on a representative mission portfolio comparable to existing NASA baselines such as the ones defined by HAT for NEAs, ESAS for the Moon, and DRA 5.0 for Mars, as outlined in the literature review in Section 1.

Table 2 shows logical constraints that have been formulated to generate this portfolio of representative missions.

Value associated with requirement sets is defined by a multiplicative multi-attribute utility (MAU) function [20], combining the following property variables:

- Total time of flight (as proxy for health risk),
- exploration time at destination,
- Delta-V capability beyond Escape orbit,
- number of crew,
- destinations,
- object size (if destination is NEA), and
- object destination (if destination is NEA).

MAU functions are elicited by interviewing expert panels. Property variables are elicited using ordinal scores. Equal complementary MAU weights have been assumed for this analysis. Not all property variables are assumed equally relevant to the three panels.

Table 3 shows the assumptions that have been used regarding relevance of each property variable for the science, exploration and policy panels that have been formed. These assumptions have been defined by an iterative verification process involving interviewees of all panels. As a working assumption, this paper only considered scenarios where all the crew would either stay in orbit or would land on a given surface destination. The analysis conducted in this paper did not therefore consider scenarios where part of the crew would stay in orbit, as means to reduce overall mass requirements of the system architecture.

4.2.5. Function-form mapping

Functions of the in-space transportation infrastructure architecture are mapped to architectural elements as shown in Fig. 2. The architecture is defined by a set of habitat elements and transportation elements. The

Table 1
Requirements Morphological Matrix.

Requirement	Requirement alternatives						No. Alt.
	1	2	3	4	5	6	
Destination	Moon	Mars	Low energy NEA	High energy NEA			4
Characteristic Destination Size	< 30 m	30–100 m	100–500 m	500–1 km	> 1 km	N/A (Dest. is not NEA)	6
Characteristic Destination Composition	Carb.	Silic.	Metallic	Other	N/A (Dest. is not NEA)		5
Number of Crew (# crew)	3	4	6				3
Exploration Time at Destination (days)	7	21	30	180	550		5
Maximum Total Mission Duration at Full-Scale Capability (years) – Health Risk Proxy	≤ 6 m	1	1.5	2	2.5	3	6
Mission Mode	Orbit	Surface					2

Table 2

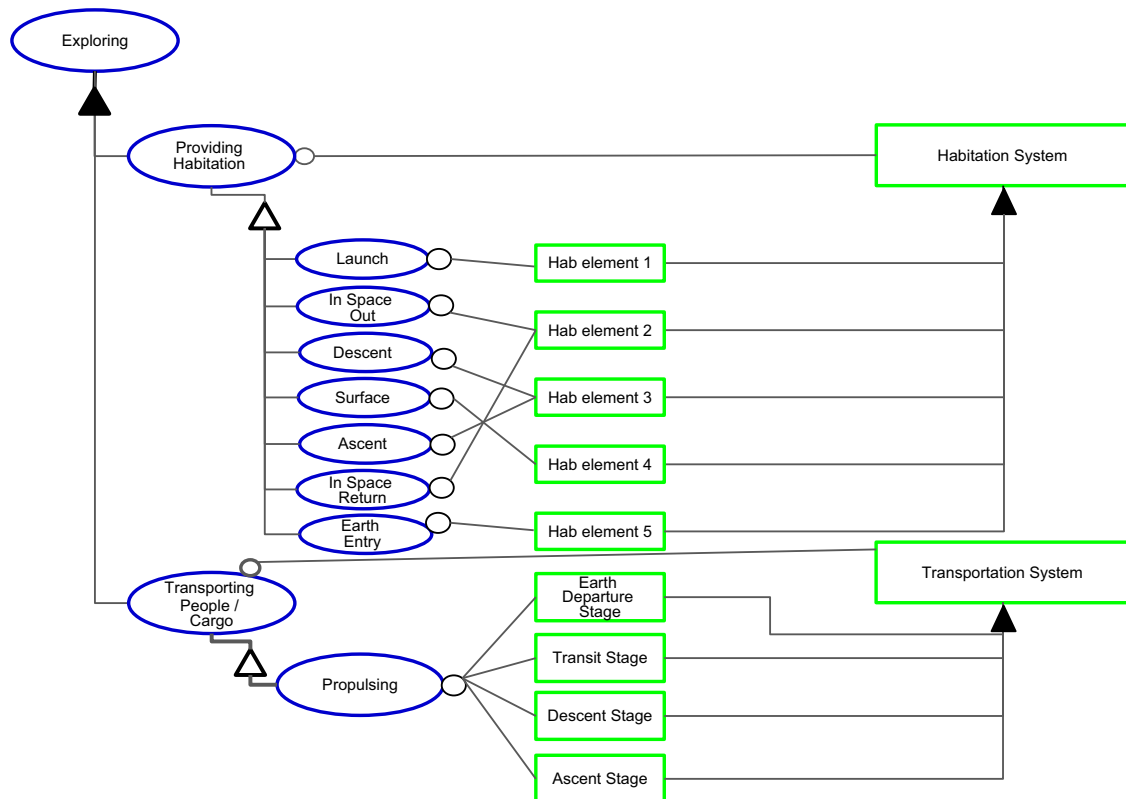
Logical Constraints for Requirements Enumeration.

Constraints Description
IF Destination is Moon, OR IF Destination is Mars, AND Characteristic Destination Size IS NOT N/A OR Characteristic Destination Composition IS NOT N/A, THEN Architecture is Infeasible
IF Destination is Mars, AND (Exploration Time is 1 m AND Total Mission Duration is NOT EQUAL TO 1 year), OR (Exploration Time is 1.5 year AND Total Mission Duration is NOT EQUAL TO 2.5 year), THEN Architecture is Infeasible
IF Maximum Total Mission Duration LESS than Exploration Time at Destination, THEN Architecture is Infeasible
IF Mission Mode is Flyby and Exploration Time at Destination IS NOT 0 day, THEN Architecture is Infeasible

Table 3

Value attributes and their assumed relevance to expert panels.

Value attributes and their relevance to expert panels	Science panel	Exploration panel	Policy panel
Total time of flight (health risk)	Less relevant	Relevant	Relevant
Exploration time at destination	Relevant	Relevant	Relevant
Delta V capability beyond escape orbit	Less relevant	Relevant	Less relevant
Number of crew	Less relevant	Relevant	Relevant
Destinations	Relevant	Relevant	Relevant
Object size (if NEA)	Relevant	Less relevant	Relevant
Object composition (if NEA)	Relevant	Less relevant	Less relevant

**Fig. 2.** Function-form mapping of the in-space transportation infrastructure.

mapping assumed here is not the only one possible for a transportation architecture. This mapping has been chosen to be compared with existing NASA point designs. Mapping functions to a varying number of elements of form can be modeled as a set partitioning problem [23],

which is an avenue of interest for future research in this application.

Different technology options for each element of form have been identified and organized in an architecting model, as described in the following discussion.

4.2.6. Architecting model

Architectural decisions and corresponding alternatives for the elements of form is organized in an architectural morphological matrix [17] as shown in Table 4.

The design space is composed by 576 possible architectures per requirement set (constrained estimate). This results into an (unconstrained) integrated design space size of ~12.4 m architectures.

The tradespace was explored with the following metrics:

- Exploration value,
- science return,
- policy return,
- Initial Mass in LEO (IMLEO), and
- Architecture Risk Ranking (ARR)

Architecture Risk Ranking (ARR) is introduced in this paper and defined as an ordinal metric to rank architectures according to programmatic risk assessment based on architectural-related (Table 5) and requirement-related (Table 6) risks. More specifically, we considered programmatic risks associated with technology selection for boiloff control, ascent/transit/TEI vehicles propellant types, and ISRU due to different flight heritage, are ranked with an ordinal scale from 1 (low risk) to 3 (high risk). Architectural risks are further distinguished by past flight heritage, as defined by proxy units of measure (such as landed mass for an aerocapture system). Architecture Risk Ranking is generally defined as a normalized weighted sum of these two risk types. This discussion presents the analysis assuming equal weights.

Table 4

Architectural morphological matrix.

Architectural decision	Architectural alternatives			No. Alt.
	1	2	3	
Cargo pre-deployment via SEP flight	Yes	No		2
Propellant boil-off control	Yes	No		2
In-situ resource utilization	Yes	No		2
Aerocapture	Yes	No		2
Trans-Departure injection propellant type	LOX/LH2	LOX/LCH4	NTR	3
Descent propellant type	LOX/LH2	LOX/LCH4		2
Ascent propellant type	LOX/LH2	LOX/LCH4		2
Trans-earth injection propellant type	LOX/LH2	LOX/LCH4	NTR	3
			Tot. Arch.	576

Table 5

Architectural Risk Elements.

Architectural risk elements					
	Proven heritage		No heritage		
Boiloff control					
Heritage unit: not relevant	N.A.	N.A.			
Yes	2	2			
No	1	1			
ISRU					
Heritage unit: not relevant	N.A.	N.A.			
Yes	3	3			
No	1	1			
Aerocapture					
Heritage unit: landed mass [mt]	N.A.	N.A.			
Yes	2	2			
No	1	1			
Transit Vehicle Propellant Type					
Heritage unit: prop. mass [mt]	≤ 120 mt	> 120 mt			
LOX/LH2	1	2			
Heritage Unit: prop. Mass [mt]	N.A.	All			
LOX/LCH4	1	2			
Heritage unit: prop. mass [mt]	≤ 75 mt	> 75 mt			
NTR	2	3			
Ascent vehicle propellant type					
Heritage unit: prop. mass [mt]	≤ 120 mt	> 120 mt			
LOX/LH2	1	2			
Heritage unit: prop. mass [mt]	N.A.	All			
LOX/LCH4	1	2			
Descent Vehicle Propellant Type					
Heritage unit: prop. mass [mt]	≤ 120 mt	> 120 mt			
LOX/LH2	1	2			
Heritage unit: prop. mass [mt]	N.A.	All			
LOX/LCH4	1	2			
TEI Vehicle Propellant Type					
Heritage unit: prop. mass [mt]	≤ 110 mt	> 110 mt			
LOX/LH2	1	2			
Heritage unit: prop. mass [mt]	N.A.	All			
LOX/LCH4	1	2			
Heritage unit: prop. mass [mt]	≤ 75 mt	> 75 mt			
NTR	2	3			

Table 6
Requirements risk elements.

Requirements risk elements			
	Risk		Risk
Destination		Time at destination	
Moon	1	0 day	1
Mars	3	7day	2
Venus	3	14day	2
NEA Low Energy	1	21day	2
NEA High Energy	1	1 m	2
		3 m	3
		6 m	3
		1 year	3
		1.5 year	3
Object Size (if NEA)		Time of flightrisk	
< 30 m	1	6 m	1
30–100 m	1	1 year	2
100–500 m	1	1.5 year	2
500–1 km	1	2.0 year	3
> 1 km	1	2.5 year	3
N/A	1	3.0 year	3
Object composition (if NEA)		Mission mode	
Carb	1	Flyby	1
Silic	1	Sortie	2
Metal	1	Surface	3
Other	1		
N/A	1		
Number of Crew		Legend	
1 Crew	3	Low risk	1
2 Crew	3	Moderate risk	2
3 Crew	2	High risk	3
4 Crew	2		
5 Crew	1		
6 Crew	1		
7 Crew	1		

4.2.7. Model validation

The model has been validated by comparison with existing architectural baselines, as shown in Table 7. The validation consisted in evaluating DRA 5.0, Apollo 11, and NASA NEA architectures with the architecting model integrating exploration/science/policy value metrics (which details are defined later in this section), a performance/cost proxy metric (defined by IMLEO), and an ordinal architectural risk metric (the ARR defined in the previous section).

This chapter integrates DB-SAF with the IMLEO estimation model for in-space infrastructure developed by [24] – who also validated IMLEO numbers using the same reference architectures used in this study – and with the novel ARR model defined in this section.

4.3. Step 3/4 – expert panel formation – problem formulation review with expert panel

Three expert panels represented Exploration, Science and Policy stakeholders. All panels were composed by senior experts involved in the architecture of the future human spaceflight infrastructure in the United States and in Europe. A total of 15 experts were involved in the study, where each expert has been engaged in a three-round DB-SAF process. Expert contributions are provided in anonymous form. Some experts agreed to be identified as participants of the study; their participation is

acknowledged at the end of the paper. In no case contributions are explicitly attributed to any of those experts to protect their anonymity. Table 8 provides a synopsis on the composition of expert panels. As per Step 4, the final problem formulation as presented in this paper has been reviewed and agreed with expert panels as part of the first iteration of the study.

4.4. Step 5 – design of interview

Interviews asked experts to give ordinal scores to value properties, such as potential exploration destinations, amount of delta-V beyond Earth Escape orbit made available by the in-space transportation infrastructure, and so forth.

4.5. Step 6 – elicitation of expert value judgment

A total of 45 interviews were conducted to inform this study. Interviews were conducted in person, on VoIP teleconference and over the phone November 2011 and January 2012.

4.6. Step 7 – results analysis

4.6.1. Exploration panel – elicitation of value functions

The exploration panel has been called to elicit value judgments on what constitutes exploration value for the in-space exploration infrastructure. Figs. 3 and 4 show DB-SAF results of the exploration panel. All DB-SAF plots can be read as follows: the red lines represent median values (50% percentile) of the answers as given by the panel. The lower and upper edges of the blue boxplots represent the 25th and 75th percentile, respectively. The black whiskers represent maximum and minimum values given by experts. Red crosses are outliers. These plots were generated at each DB-SAF iteration and presented to experts. Experts could see where their answers lay in the distributions, and could change them to either move toward consensus (median value), diverge from consensus, or keep their answers through iterations. In all cases, participants were asked to provide justification for their positions. By interacting between each other in anonymous form through DB-SAF, experts were able to express their judgments without peer pressure and being able to explain their reasoning clearly, therefore giving the opportunity to document the whole process for analysis. Experts in the exploration panel reached the best consensus among panels. The figure shows six plots, each one representing one value metric. The sixth panel shows weights assigned to value metrics in the aggregate multiplicative (hence nonlinear) MAU formulation. The convention in all value metrics – in this and on science and policy panels as well – is that of *value-maximizing stakeholders*.

Time of Flight (as a proxy for health risk): Time of flight is a proxy of health risk as astronauts are more likely to face adverse conditions from exposure to the harsh radiation environment beyond the Van Allen Belts. The shape and gradient on this curve depend on risk aversion of individual experts in the panel. Risk-averse experts argue that safety is a key feature for any future mission of the human

Table 7
Model Validation.

Low Energy NEA (HAT 2000SG34)		High Energy NEA (HAT 2008EV5)	
Destination	NEA	Destination	NEA
Characteristic Size	300 m	Characteristic size	300 m
Characteristic Composition	Carbonaceous	Characteristic composition	Carbonaceous
Number of Crew	4	Number of Crew:	4
Exploration Time at Destination	~7 days	Exploration time at destination	~30 days
Total Mission Duration	~1 year	Total Mission Duration:	~1.5 years
Mission Mode	Surface	Mission mode	Surface
Boiloff Control	No	Boiloff contro:	No
ISRU	No	ISRU	No
Aerocapture	N/A	Aerocapture:	N/A
Transit prop. type	LOX/LH2	Transit prop. type	LOX/LH2
Ascent prop. type	N/A	Ascent prop. type	N/A
Descent prop. type	N/A	Descent prop. type	N/A
TEI prop. type	NTO/N2O4 (MPCV)	TEI prop. type	NTO/N2O4 (MPCV)+LOX/LH2
Transit vehicle heritage	No heritage	Transit vehicle heritage	No heritage
Ascent vehicle heritage	No heritage	Ascent vehicle heritage	No heritage
Descent vehicle heritage	No heritage	Descent vehicle heritage	No Heritage
TEI vehicle heritage	No heritage	TEI vehicle heritage	No heritage
Total IMLEO:	2 × 70 mt SLS (140 mt)	Total IMLEO:	3 × 70 mt SLS (210 mt)
Model-generated evaluations		Model-generated evaluations	
Exploration value:	0.35	Exploration value:	0.65
Science value:	0.5	Science value:	0.68
Policy value:	0.33	Policy value:	0.69
Architectural Risk Ranking:	0.33	Architectural Risk Ranking:	0.33
NASA Design Reference Architecture 5.0		Apollo 11	
Destination	Mars	Destination:	Moon
Characteristic size	N/A	Characteristic size	N/A
Characteristic composition	N/A	Characteristic composition	N/A
Number of Crew:	6	Number of crew	3
Exploration time at destination	~1.5 years	Exploration time at destination	~3 days
Total Mission Duration	~2.5 years	Total mission duration	~8 days
Mission mode	Surface	Mission mode	Surface
Boiloff control	Yes	Boiloff control	No
ISRU	Yes	ISRU:	No
Aerocapture	No	Aerocapture	N/A
Transit prop. type	NTR/LH2	Transit prop. type	LOX/LH2
Ascent prop. type:	LOX/LCH4	Ascent prop. type	NTO/N2O4
Descent prop. type:	LOX/LCH4	Descent prop. type	NTO/N2O4
TEI Prop. Type:	NTR/LH2	TEI prop. type	NTO/N2O4
Transit vehicle heritage	No Heritage	Transit vehicle heritage	Proven Heritage
Ascent vehicle heritage	No heritage	Ascent vehicle heritage	Proven heritage
Descent vehicle heritage	No heritage	Descent vehicle heritage	Proven heritage
TEI vehicle heritage	No heritage	TEI vehicle heritage	Proven heritage
Total IMLEO:	848.7 mt	Total IMLEO:	1 × Saturn V (120 mt)
Model-generated evaluations		Model-generated evaluations	
Exploration value:	1.00	Exploration value:	0.24
Science value:	1.00	Science value:	0.28
Policy value:	0.81	Policy value:	0.15
Architectural Risk Ranking:	0.88	Architectural Risk Ranking:	0.21

spaceflight program. Risk-taker experts point at an excessive stress to safety as one of the main causes of cost growth and schedule slippage in past exploration programs. They argue that exploration as such is an inherently risky venture. Therefore, risk should be accepted as part of the program to enable sustainable access to new destinations in the Solar System.

Time at destination (Moon/Mars): experts in the exploration panel were able to reach consensus on value associated with surface stay times on large planetary bodies. The underlying curve is an S-curve, where the shape and the gradient are representative of the marginal value increase or saturation associated with short stay and

long stay durations. Experts agree that value associated with surface stay time has increasing marginal returns for short stays, as contribution to exploration value in the first days of exploration is significant, as new areas are explored and new operations initiated on the surface. The right-hand side of the curve shows diminishing marginal returns, meaning that the added value of an additional day on the surface when several months have already been spent there is low. Experts were specifically asked not to include orbital constraints in their assessment. This was motivated by the intent to derive a curve to compare propulsion technologies and mission modes in a second phase of the analysis.

Table 8
Expert Panels Composition.

Exploration panel
4 Senior Systems Architects & Decision Makers from NASA and ESA
1 Professor from Academia (US)
Science panel
3 Professors from Academia (US)
1 Senior Scientist from a US Research Institution
1 Senior Scientist from a European Industrial Organization
Policy panel
2 Decision Makers from NASA and ESTEC
1 Senior Expert (former President of a EU Space Agency)
1 Senior Advisor from a US Research Institution
1 Professor from Academia (US)

Time at destination (NEA): time at NEA destinations was measured on a different time scale as agreed on by experts. The open debate in this context can be traced to different assumptions by experts on time required for preparatory and system check-out operations. Such ambiguity could not be reduced in three rounds of iterations among experts, despite sharing expert opinions on those matters. Further discussion and drafting of a specification in this context with particular reference to detailed concepts of operations in the public domain of a NEA mission are advisable to reduce this ambiguity.

Number of crew: experts reached consensus in that 3 (same crew size as Apollo 11) is the minimum number of crew required for any exploration beyond LEO. Marginal value returns with added crew members is mostly linear as being proportional to the cumulative number of exploration operations that could be conducted on the mission. Diminishing marginal returns begin at a crew size of 4. This coheres with the current baseline of the Multi-Purpose Crew Exploration Vehicle (MPCV) as developed by NASA [25].

Relative weights: experts agree that the choice of a destination for exploration and time spent at the destination are primary drivers in the evaluation of exploration architecture. Relative ranking between the two is unclear, as the two boxplots mostly overlap, signaling a tie in this context. Number of crew is the second most relevant attribute from an exploration perspective. Time of flight is predominantly seen as the last factor of interest in evaluating architectures.

Destinations: the choice of a destination shows the most irreducible ambiguity in all panels that have been involved in the study. The debate on destinations is found to be strongly opinionated – while in previous decisions experts were willing to compromise, DB-SAF iterations on destinations showed little room for negotiations. While evolutionary patterns between fly-by, orbit, sortie and long stay mission modes were widely recognized across the panel, the relative ranking between destinations was not. Furthermore, some destinations generated interest only with some experts – such as EM-L1 or Venus. Notably, NEAs are part of this category. While NASA's current efforts are focused on the development of a mission to a NEA, not all experts agree on the validity of such destination.

4.6.2. Exploration panel – main highlights

The following points emerge as main highlights from the development of representative value functions for the exploration panel.

4.6.2.1. Time at destination (NEA)

- Exploration value between 14-days and 21-days is an irreducible ambiguity in different assumptions on Concept of Operations (CONOPS) – in particular due to different assumptions on time required for system check-out and testing before commitment to NEA surface operations.
- The development of a common and open standard on a detailed mission CONOPS for NEA missions is recommended.

4.6.2.2. Destinations

- Highest degree of irreducible ambiguity. At the same time, the panel ranked destinations as first most important attribute in the evaluation procedure.
- Debate is highly opinionated, impossibility of reaching consensus.
- Open debate on destination selection is highly recommended to reduce associated ambiguity within NASA and with international partners. Of particular interest is reduction of ambiguity on NEA mission architectures, which evaluation in terms of exploration panel appears bimodal, with experts giving high value to NEA missions versus experts giving zero value to NEA missions. Both parties did not agree in modifying their position at any point during the debate due to fundamental different value assumptions.

4.6.3. Science panel – elicitation of value functions

Science experts shared the opinion that science in a human spaceflight program is a complementary objective – certainly important but not primary. DB-SAF results of the science panel are shown in Figs. 5 and 6. A discussion on the set of property variables of interest to scientists follows.

Time at destination (Moon/Mars): Science return is perceived as proportional to exploration time at the destination.

Time at destination (NEA): Value associated with time on NEAs featured a double plateau, due to different types of science that could be done on that mission. Remote sensing and internal tomography of NEA interiors is possible via orbit operations, therefore representing early value delivering activities such a science program. Additional time spent (associated with NEA surface operations) increases value significantly by enabling sample return and geological field work activities. A maximum time of 3 months is foreseen on the surface for a first NEA mission.

Object size (NEA): for NEAs with a characteristic size greater than 30 m, science value is insensitive to object size.

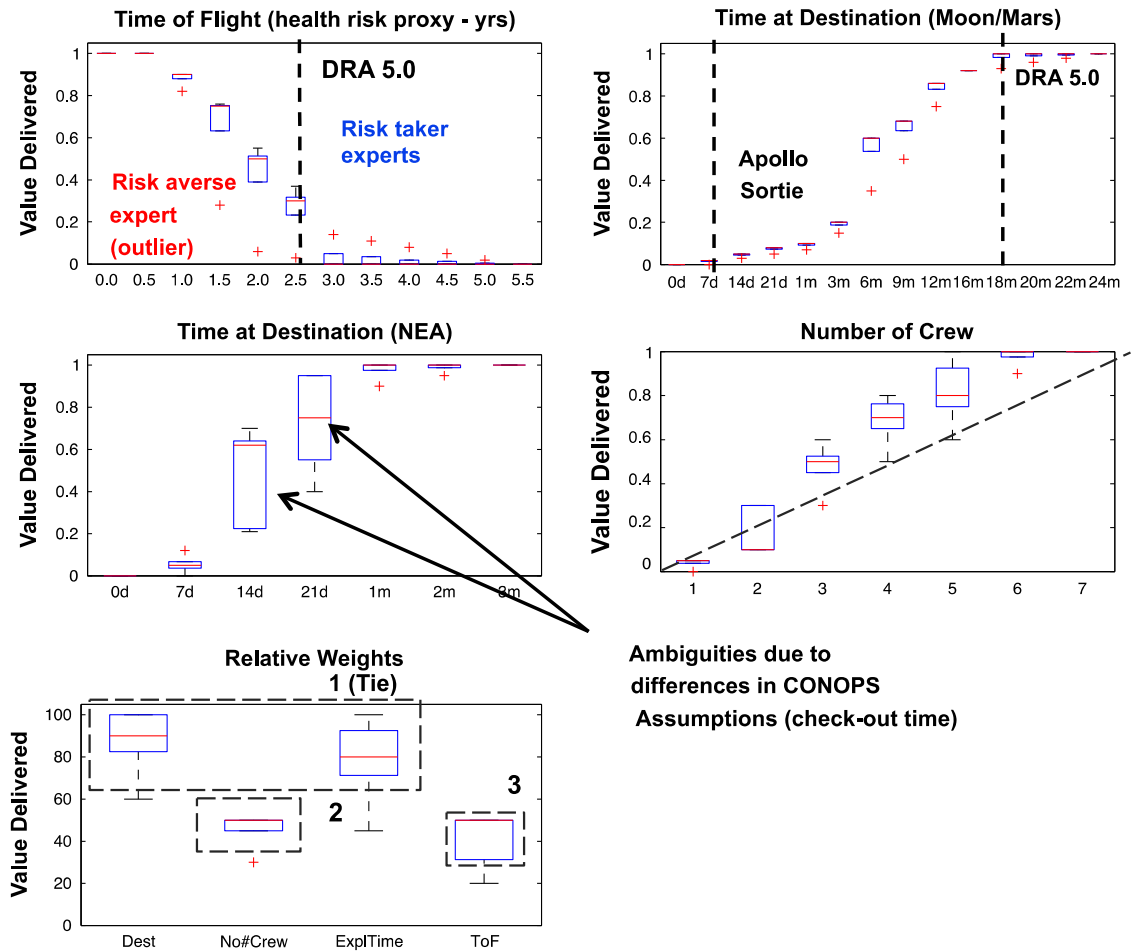


Fig. 3. Exploration panel – round 3. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

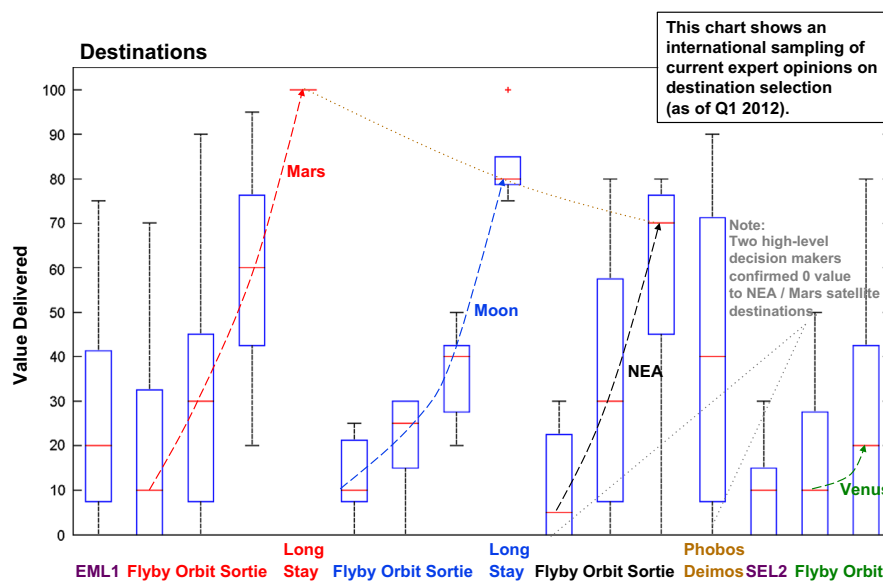


Fig. 4. Exploration panel – destination choice – round 3. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

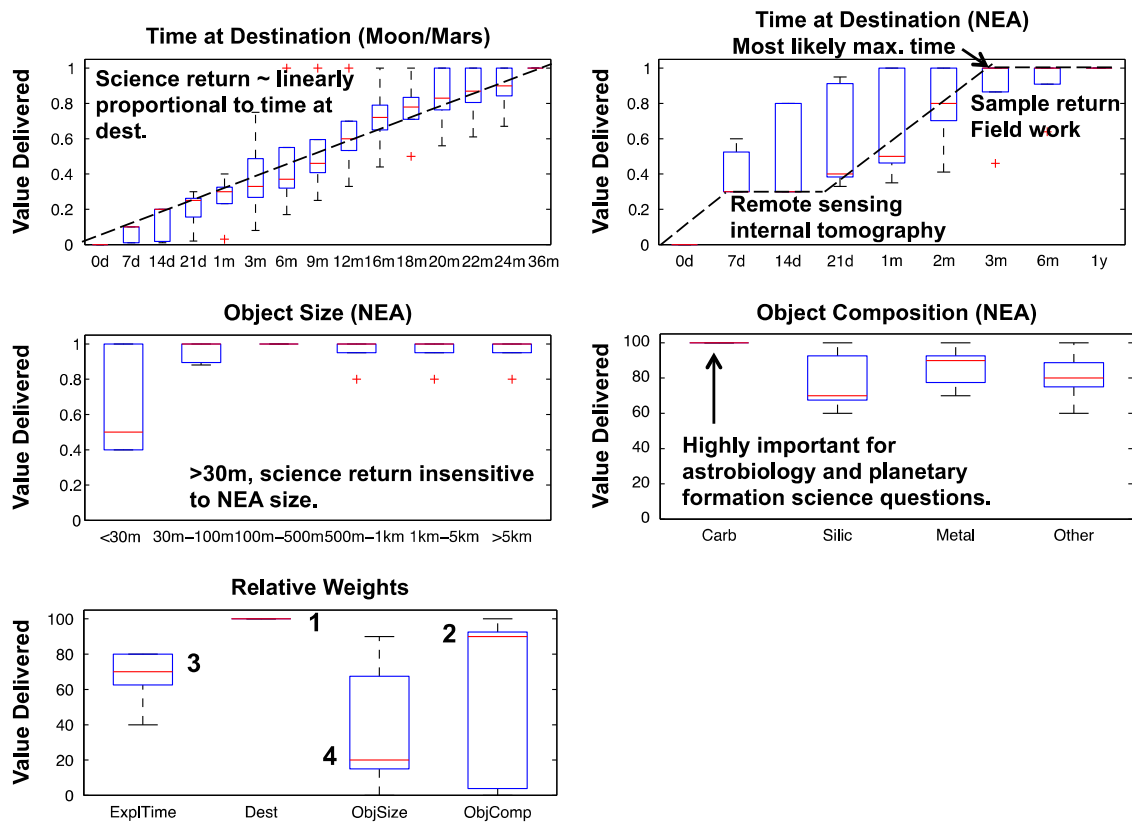


Fig. 5. Science panel results – Round 3.

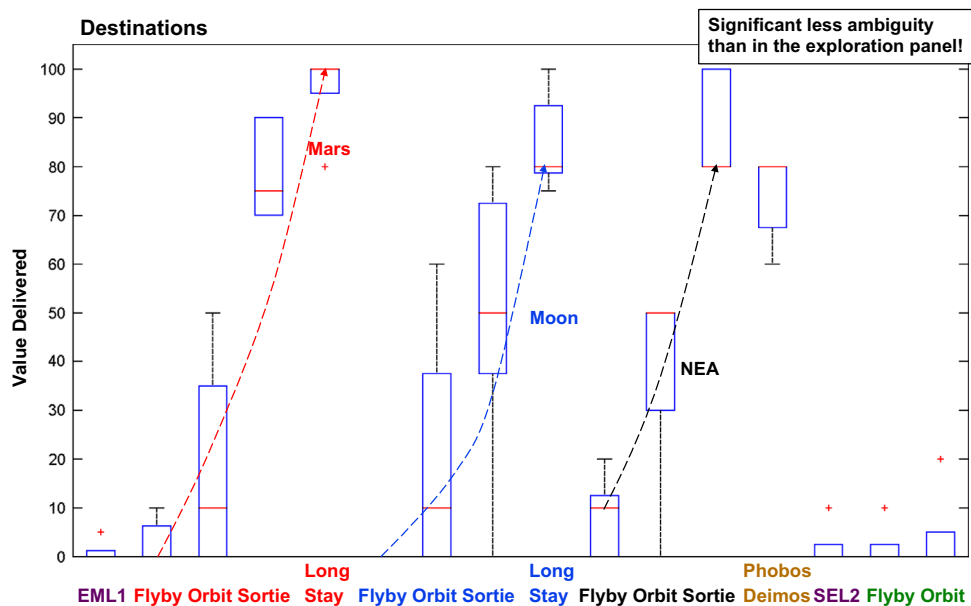


Fig. 6. Science panel – destination choice – Round 3.

Object composition (NEA): While all compositions are of interest to answer questions in planetary science, carbonaceous objects emerge as clear priorities due to their importance in answering relevant questions in astrobiology and planetary formation. Carbonaceous NEAs are

samples of thermally unprocessed, pristine material from the origin of the Solar System.

Relative weights: the science panel followed a value-driven approach, where the choice of a destination was clearly recognized as the first priority in evaluation.

Second to it object composition, although this opinion varied according to whether the expert in question was concerned with science related to NEA composition (such as geologists). Time spent at the destination emerged as third priority. Object size and time of flight tied as last priorities, as also shown in the discussion above.

Destinations: Clear evolutionary trends are found in Mars, Moon and NEA missions. As NEAs are valuable evidence to study planetary formation and the origin of the Solar System, they were retained by the panel of high science value. No or little science value was associated to EM-L1, SE-L2 and Venus manned flyby/orbit missions, as neither of the Venus missions benefit from humans as enablers of science.

4.6.4. Science panel – main Highlights

- Humans are seen as enablers of certain science (such as sample return and geology field work) and evaluations calibrated accordingly.
- Carbonaceous asteroids are a privileged NEA category as they are samples of thermally unprocessed, pristine material answering questions on planetary formation and the history of the Solar System.
- Object composition is key for NEA mission selection. NEA size does not matter, unless looking at specific NEA science questions such as rubble pile versus onion shell theories for NEA internal composition, which can also be answered by use of tomography instruments.

4.6.5. Policy panel – elicitation of value functions

The policy panel followed more qualitative logics in the evaluation than other panels. DB-SAF policy panel results are shown in Figs. 7 and 8.

Total time of flight (as proxy for health risk): time of flight was not seen only in light of technical constraints, but also considering policy constraints such as short-term policy return. Certain policy experts in the panel motivated their answers based on the fact that the US Congress changes in 2 year cycles and a US President in 4 year (one term) and 8 year (two terms) cycles. Ambiguity on this value function could be effectively reduced in three iterations.

Time at Destination (Moon): the panel proposed to consider this measure separately for Moon, Mars and NEAs. For the Moon, the maximum was identified in 6 months for a lunar base settlement, in agreement with previous experience with the International Space Station. The trend over time follows a curve with diminishing marginal returns. Value was estimated as zero for a seven day Moon sortie, as experts stated the desire for new Moon programs to advance from the Apollo era and have an impact on media and the public opinion.

Time at Destination (Mars): a step-wise trend was identified by the panel, marking the difference between 30 day sortie and 550 sortie mission modes. The DRA 5.0 baseline for surface stay (~18 months) is on the point of diminishing marginal returns of the curve.

Time at Destination (NEA): a clear trend showing diminishing marginal returns was identified by the panel. Value saturates at 1 month of exploration, with substantial value

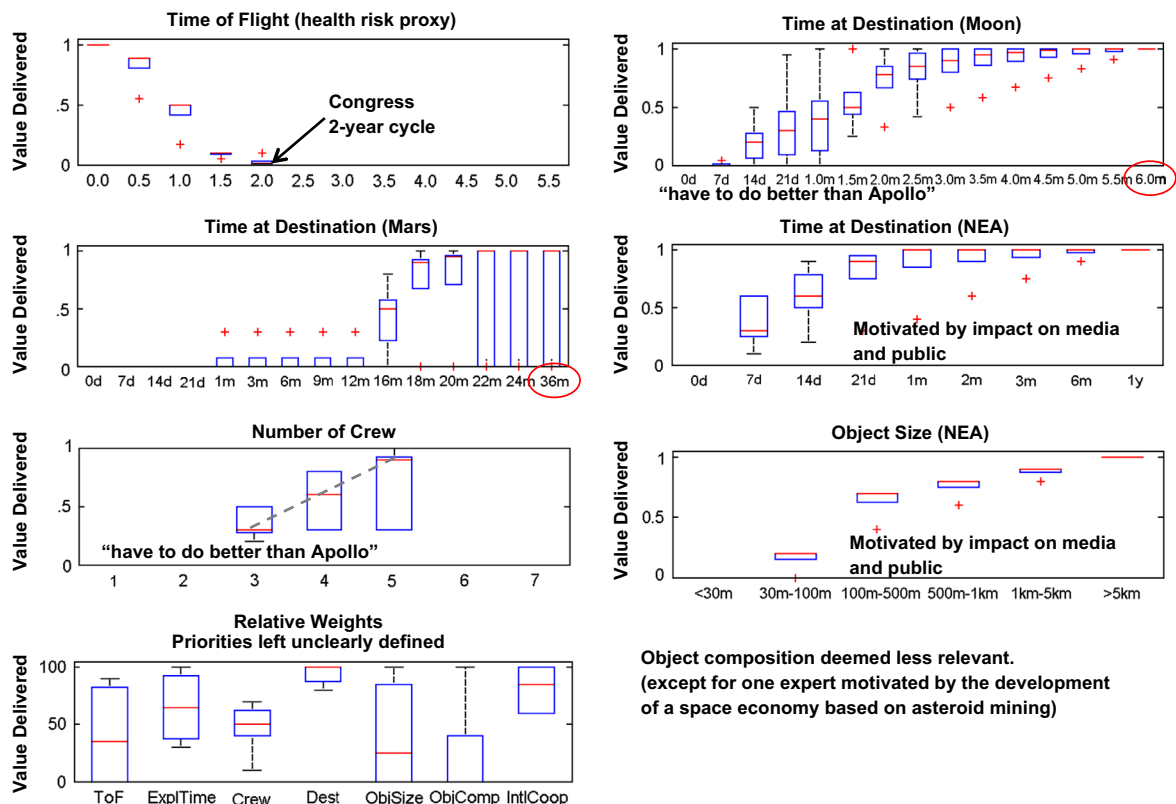


Fig. 7. Policy panel – Round 3.

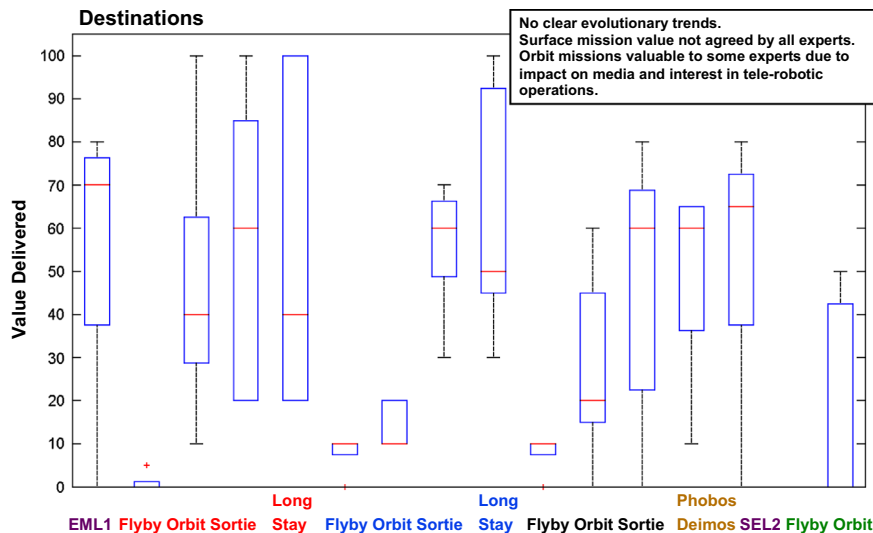


Fig. 8. Policy panel – destination choice – Round 3.

delivery (>50%) at 2 weeks of exploration. This was motivated as policy experts believe such a mission would have greater impact on media and the public in the short term rather than in the long term. This conclusion suggests keeping mission duration in the order of one month, in accordance with NASA baseline NEA architectures.

Number of Crew: the panel took the Apollo program as a reference considering a crew range between 3 and 6 crew members, with a linear increase in value in this range.

Object Size: value increases proportionally to object size showing diminishing marginal returns. Object size is related to the impact on the media and the public associated with the mission.

Object composition was deemed irrelevant by policy experts. An exception is made by experts advocating for the development of a space economy, who are therefore interested in NEA objects which composition suits the needs for the development of asteroid mining ventures.

Relative Weights: an unclear priority ranking was elicited by the panel. High irreducible ambiguity is due to the plurality of non reconcilable intents.

This last result implies the lack of strong leadership driving worldwide policy opinions.

Destinations: No clear evolutionary trends were identified between missions modes (flyby, orbit, sortie and long stay). In some cases, orbit missions were deemed more valuable than surface missions as experts believed these alone deliver significant impact to the media and on public opinion, while enabling exploration opportunities provided by tele-robotic exploration on planetary surfaces – therefore mitigating communication lags.

4.6.6. Policy panel – main highlights

- No clear leadership or consensus emerges, which implies the need to bring concepts for further evaluation.
- Time scale arguments were based on impact on media and public opinion, and were based on coherence with

possible political changes rather than scientific/engineering rationale.

- No evolutionary trends in destination choice were found between mission modes (flyby, orbit, sortie and surface long stay) as perceived by other panels.
- Orbit missions were deemed valuable by some policy experts for their impact on media and public opinion (such as in Apollo 8).

4.7. Step 8 – aggregate results discussion

The expert elicitation process described so far represents the first part of the study, where system objectives for the in-space infrastructures have been identified along with their associated uncertainties – or ambiguities. In other words, the study identified how value – or benefit – is measured by stakeholders for a transportation infrastructure. Leveraging on this result, this section shows a first-order benefit-cost analysis of the in-space transportation infrastructure, and performs tradespace exploration to characterize architectural decisions.

The proxy measure for benefit is defined as the weighted average of median values of value functions derived by the expert panels. Weights represent panel preferences. A proxy for cost is defined as the total delta-V that the infrastructure needs to provide for a given destination/mission mode (for instance Moon Sortie or Mars Long Stay). We neglect Venus in this analysis, as expert elicitation results have shown this choice was dominated in value in all panels by other destinations.

Sensitivity analysis results to different panel weights are shown in Figs. 9–12 in the Appendix. The following results emerge from the first-order benefit/cost analysis shown in the abovementioned figures:

- NEA destinations are always on the Pareto front. Low-energy NEAs represent efficient benefit/cost tradeoffs being on the kick of the curve. This result is confirmed in

all the scenarios considered in the sensitivity analysis.

- While high-energy NEAs are always dominated in a Pareto sense, they represent destinations of interest as they require higher capabilities in terms of delta V, better approaching energy requirements of future Mars missions.
- Value associated with mission duration and mission mode (flyby vs. orbit vs. sortie vs. long stay) is moderately panel dependent. Said value is higher for exploration and science biased scenarios, for which longer durations are preferred. Value is lower in the policy-biased scenario, as policy experts showed preference for shorter mission durations for early value return within Presidential and Congressional mandates.
- Value associated with EM-L1 and SE-L2 destinations is highly panel dependent. They show high value (close to the Pareto front) in the policy-biased scenario, as these destinations comply with current policy guidance (as of Q1 2012), and represent a gateway for future exploration as discussed previously. An exploration-biased assumption returns moderate value from these destinations, since technology development is enabled by these destinations (while having lower exploration value when compared to other alternatives). Value is low in a science-bias assumption, as EM-L1 and SE-L2 are empty points in deep space, where no human-enabled science (such as sample return) can be performed.

4.8. Step 9 – convergence criteria

A maximum number of three iterations has been used as termination criteria for DB-SAF. The choice of a convergence criteria resulted from a tradeoff between expert's

time availability and the focus of the architecting study. Convergence has been verified with experts; they confirmed that they were unlikely to change their answers in a fourth round of DB-SAF, therefore the study was called to a close.

4.9. Step 10 – recommendations

The following recommendations emerge from the analysis: (1) The choice of a destination is the predominant irreducible ambiguity affecting the architecture of in-space transportation infrastructure, with particular impact on exploration-related and policy-related stakeholders. (2) No clear leadership or consensus emerging among policy makers in setting a direction for human spaceflight, which implies the need for NASA to bring new concepts for further evaluation. (3) In NEA architectures, prefer carbonaceous, large-sized (> 100 m) NEA destinations to leverage on added value of human presence for enhanced sample return. (4) Develop an open standard on a detailed mission CONOPS for NEA missions to reduce ambiguity on exploration value definition.

5. Summary and conclusions

This paper implemented a Delphi-Based Systems Architecting Framework for the architecting problem of in-space transportation infrastructure for human exploration beyond Low-Earth Orbit. It identified ambiguities in the definition of the functional intent in the infrastructure architecture, with direct impact on value delivery to stakeholders as expressed by goals. The most important result produced by this analysis is the identification of a critical irreducible ambiguity undermining success in delivering value for the

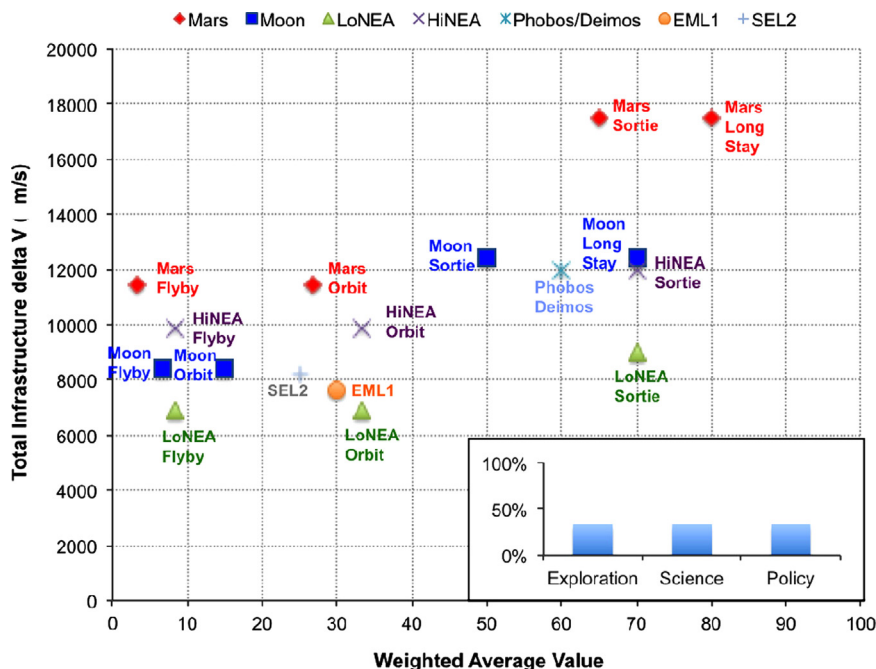


Fig. 9. Benefit-cost analysis results – equal weights (33.33% exploration, 33.33% science, 33.33% policy).

in-space transportation infrastructure of the next three decades: destination choice. While NASA's program of record at the time of writing of this paper is focused on architecting a human spaceflight mission to a Near Earth Asteroid, consensus on this architecture is far from being reached by the community at large, with particular reference to exploration and policy stakeholders.

The realization of this ambiguity is a call for NASA to promote an open forum on this topic, and to develop a strong case for policy makers to incentivize investments in the human spaceflight industry in the next decades. Ambiguity and current lack of consensus in policy leadership, as it emerged in this study, are critical issues to be addressed in today's early design stages of the design of

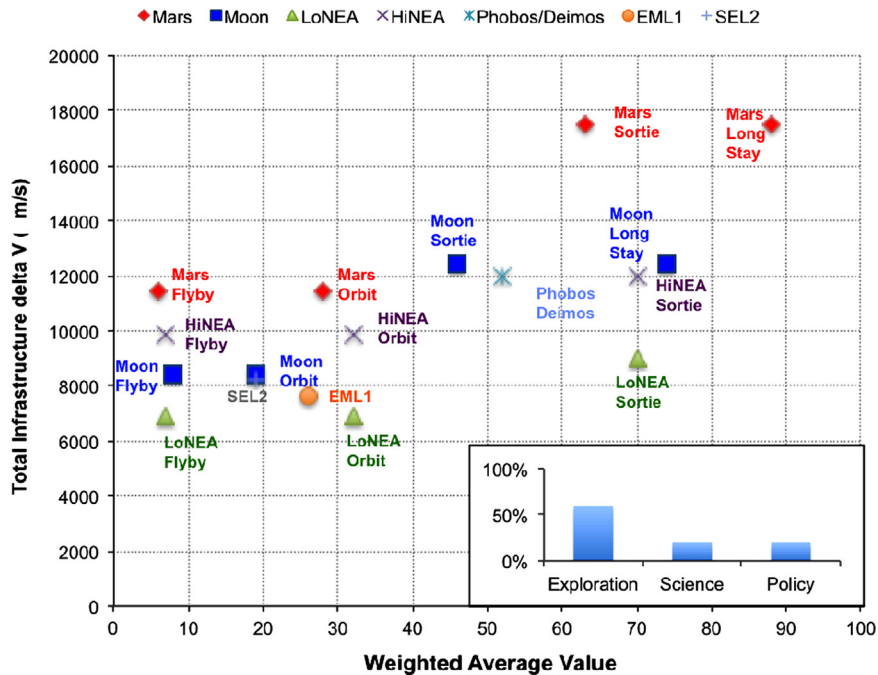


Fig. 10. Benefit-cost analysis results – exploration bias scenario (60% exploration, 20% science, 20% policy).

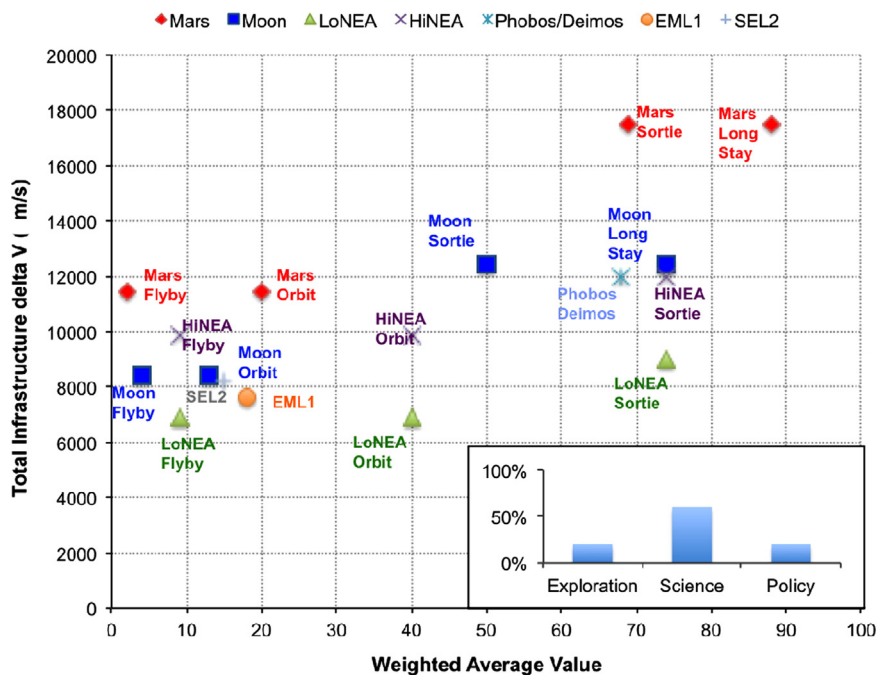


Fig. 11. Benefit-cost analysis results – science bias scenario (20% exploration, 60% science, 20% policy).

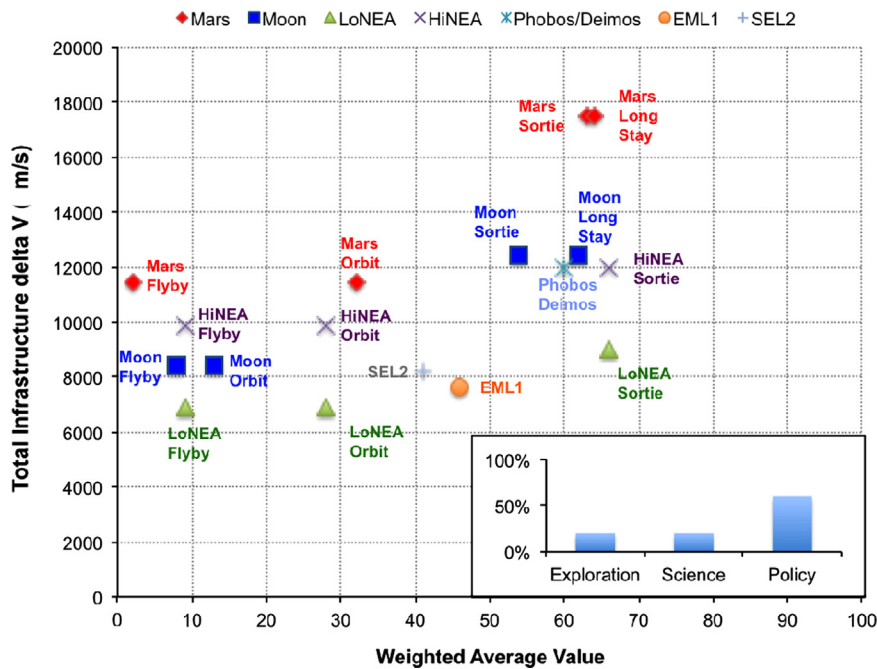


Fig. 12. Benefit-cost analysis results – policy bias scenario (20% exploration, 20% science, 60% policy).

in-space systems for successful development, implementation, operations and value delivery of the space infrastructure of tomorrow.

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Appendix

See Figs. 9–12.

References

- [1] A. Aliakbargolkar, A.C. Wicht, J.A. Battat, E.D. Calandrelli, E.F. Crawley, Heavy Lift Launch Vehicle Systems Architecting, in: Proceedings of the 62nd International Astronautical Congress, International Astronautical Federation, Cape Town, South Africa, 2011.
- [2] K. Chang, NASA Unveils New Rocket Design, in: New York Times, New York, 2011.
- [3] A. Aliakbargolkar, A framework for space systems architecting under stakeholder objectives ambiguity, Aeronautics and Astronautics, Massachusetts Institute of Technology, Cambridge, MA, 2012.
- [4] J.M. Logsdon, A sustainable rationale for human spaceflight, Issues Sci. Technol. 20 (2004) 31–34.
- [5] R.D. Launius, Compelling Rationales for Spaceflight? History and the Search for Relevance, Critical Issues in the History of Spaceflight, NASA, Washington 37–70.
- [6] D.A. Mindell et al., The Future of Human Spaceflight, Space, Policy, and Society Research Group, Massachusetts Institute of Technology, December 2008.
- [7] NASA, The Vision for Space Exploration, 2004.
- [8] K. Chang, Obama calls for end to NASA's moon program, New York Times, The New York Times Company, New York, 2010.
- [9] NASA, Review of US Human Spaceflight Plans Committee – Final Report, 2009.
- [10] J.F. Connolly, Constellation Program Overview, in: Proceedings of NASA Presentation, 2006.
- [11] K. Chang, The shuttle ends its final voyage and an era in space, The New York Times, The New York Times Company, New York, 2011.
- [12] NASA, NASA Announces Design for New Deep Space Exploration System, in: NASA, 2011.
- [13] G. Rowe, G. Wright, The Delphi technique as a forecasting tool: issues and analysis, Int. J. Forecast. 15 (1999) 353–375.
- [14] G. Rowe, G. Wright, F. Bolger, Delphi: a reevaluation of research and theory, Technol. Forecast. Soc. (1991) 235–251.
- [15] H.Y.B. Koo, A Meta-Language for Systems Architecting, Massachusetts Institute of Technology, 2005.
- [16] W.L. Simmons, A Framework for Decision Support in Systems Architecting, Massachusetts Institute of Technology, 2008.
- [17] M.I.T. Draper, Concept Exploration and Refinement – Final Report, Massachusetts Institute of Technology, Cambridge, MA, 2005.
- [18] NASA, Exploration Systems Architecture Study – Final Report, in, 2005.
- [19] B.G. Drake, in: M.A.S. Group (Ed.), Human Exploration of Mars – Design Reference Architecture 5.0, NASA Headquarters, Washington DC, 2009.
- [20] NASA, Human Exploration Framework Team (HEFT), 2010.
- [21] C. Culbert, Human Space Flight Architecture Team (HAT) Overview, in: Proceedings of GER Workshop, 2011.

- [22] ISECG, Global Exploration Roadmap, in: ISECG (Ed.), 2011.
- [23] D. Selva, Rule-based System Architecting of Earth Observation Satellite Systems, Aero/Astro, Massachusetts Institute of Technology, Cambridge, MA, 2012.
- [24] A. Rudat, J.A. Battat, A. Aliakbargolkar, M. Dwyer, B.G. Cameron, E.F. Crawley, Tradespace exploration approach for architectural definition of in-space transportation infrastructure systems for future human space exploration, in: Proceedings of the International Astronautical Congress IAC 2012, International Astronautical Federation, Naples, Italy, 2012.
- [25] NASA, Orion Multi-Purpose Crew Vehicle, 2012.